ADV: Logs and Exponentials (Adv), E1 Logs and Exponentials (Adv)

Graphs and Applications (Y11)
Log/Index Laws and Equations (Y11)

Teacher: Ms Najem

Exam Equivalent Time: 129 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

Questions

1. Functions, 2ADV' F1 2019 HSC 1 MC

What is the domain of the function $f(x) = \ln(4 - x)$?

A. $x < 4$

B. $x \leq 4$

C. $x > 4$

D. $x \geq 4$

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2. L&E, 2ADV E1 SM-Bank 2 MC

If $y = \log_a(7x - b) + 3$, then x is equal to

(A) $\dfrac{1}{7}a^{y-3} + b$

(B) $\dfrac{y-3}{\log_a(7-b)}$

(C) $\dfrac{1}{7}\left(a^{y-3} + b\right)$

(D) $a^{y-3} - \dfrac{b}{7}$

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3. L&E, 2ADV E1 SM-Bank 4 MC

If $f(x) = 3\log_e(2x)$, and $f(5x) = \log_e(y)$,

then y is equal to

(A) $30x$

(B) $6x$

(C) $125x^3$

(D) $1000x^3$

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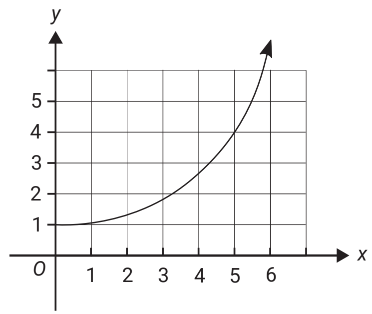
4. Algebra, STD2 A4 SM-Bank 1 MC

The graph below represents $y = a^x$.



What is the approximate value of $2\left(a^5\right)$?

A. 4

B. 8

C. 32

D. 64

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5. L&E, 2ADV E1 2012 HSC 7 MC

Let $a = e^x$

Which expression is equal to $\log_e\left(a^2\right)$?

(A) e^{2x}

(B) e^{x^2}

(C) $2x$

(D) x^2

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6. L&E, 2ADV E1 2013 HSC 9 MC

What is the solution of $5^x = 4$?

(A) $x = \dfrac{\log_e 4}{5}$

(B) $x = \dfrac{4}{\log_e 5}$

(C) $x = \dfrac{\log_e 4}{\log_e 5}$

(D) $x = \log_e\left(\dfrac{4}{5}\right)$

7. L&E, 2ADV E1 2014 HSC 3 MC

What is the solution to the equation $\log_2(x-1) = 8$?

(A) 4

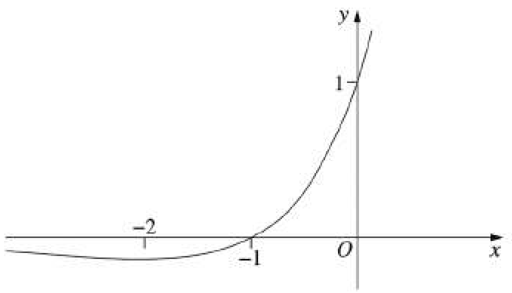
(B) 17

(C) 65

(D) 257

8. L&E, 2ADV E1 2015 HSC 8 MC

The diagram shows the graph of $y = e^x(1+x)$.



How many solutions are there to the equation $e^x(1+x) = 1 - x^2$?

(A) 0

(B) 1

(C) 2

(D) 3

9. L&E, 2ADV E1 2017 HSC 5 MC

It is given that $\ln a = \ln b - \ln c$, where $a, b, c > 0$.

Which statement is true?

(A) $a = b - c$

(B) $a = \dfrac{b}{c}$

(C) $\ln a = \dfrac{b}{c}$

(D) $\ln a = \dfrac{\ln b}{\ln c}$

10. L&E, 2ADV E1 2019 HSC 3 MC

What is the value of p so that $\dfrac{a^2 a^{-3}}{\sqrt{a}} = a^p$?

(A) -3

(B) $-\dfrac{3}{2}$

(C) $-\dfrac{1}{2}$

(D) 12

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11. L&E, 2ADV E1 2019 HSC 5 MC

Which of the following is equal to $\dfrac{\log_2 9}{\log_2 3}$?

(A) 2

(B) 3

(C) $\log_2 3$

(D) $\log_2 6$

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12. L&E, 2ADV E1 SM-Bank 11 MC

Solve the equation $e^{4x} - 5e^{2x} + 4 = 0$ for x

(A) $x = 1, 4$

(B) $x = -4, -1$

(C) $x = 0, \log_e 2$

(D) $x = -\log_e 2, 0, \log_e 2$

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13. L&E, 2ADV E1 SM-Bank 6 MC

The expression

$$\log_c(a) + \log_a(b) + \log_b(c)$$

is equal to

(A) $\dfrac{1}{\log_c(a)} + \dfrac{1}{\log_a(b)} + \dfrac{1}{\log_b(c)}$

(B) $\dfrac{1}{\log_a(c)} + \dfrac{1}{\log_b(a)} + \dfrac{1}{\log_c(b)}$

(C) $-\dfrac{1}{\log_a(b)} - \dfrac{1}{\log_b(c)} - \dfrac{1}{\log_c(a)}$

(D) $\dfrac{1}{\log_a(a)} + \dfrac{1}{\log_b(b)} + \dfrac{1}{\log_c(c)}$

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14. L&E, 2ADV E1 2006 HSC 1a

Evaluate $e^{-0.5}$ correct to three decimal places. ***(2 marks)***

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15. Functions, 2ADV F1 SM-Bank 53

i. If $\dfrac{1}{\sqrt[3]{7+\pi}} = (7+\pi)^x$, find x. ***(1 mark)***

ii. Calculate the value of $\dfrac{1}{\sqrt[3]{7+\pi}}$ to 3 significant figures. ***(1 mark)***

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16. L&E, 2ADV E1 SM-Bank 8

Solve the equation $3^{-4x} = 9^{6-x}$ for x. ***(2 marks)***

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17. L&E, 2ADV E1 SM-Bank 12

Solve the equation $\log_e(3x+5) + \log_e(2) = 2$, for x. ***(2 marks)***

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18. L&E, 2ADV E1 2011 HSC 1c

Solve $2^{2x+1} = 32$. ***(2 marks)***

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19. L&E, 2ADV E1 SM-Bank 2

The population of Indian Myna birds in a suburb can be described by the exponential fuction

$$N = 35e^{0.07t}$$

where t is the time in months.

i. What will be the population after 2 years? *(1 mark)*

ii. Draw a graph of the population. *(2 marks)*

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20. L&E, 2ADV E1 2005 HSC 5a

Use the change of base formula to evaluate $\log_3 7$, correct to two decimal places. *(1 mark)*

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21. Functions, 2ADV F2 SM-Bank 1

i. Draw the graph $y = \ln x$. *(1 mark)*

ii. Explain how the above graph can be transformed to produce the graph

$$y = 3\ln(x+2)$$

and sketch the graph, clearly identifying all intercepts. *(3 marks)*

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22. L&E, 2ADV E1 SM-Bank 14

The spread of a highly contagious virus can be modelled by the function

$$f(x) = \frac{8000}{1 + 1000e^{-0.12x}}$$

Where x is the number of days after the first case of sickness due to the virus is diagnosed and $f(x)$ is the total number of people who are infected by the virus in the first x days.

i. Calculate $f(0)$. *(1 mark)*

ii. Find the value of $f(365)$ and interpret it result. *(2 marks)*

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23. L&E, 2ADV E1 SM-Bank 7

Solve the equation $2\log_3(5) - \log_3(2) + \log_3(x) = 2$ for x. *(2 marks)*

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24. L&E, 2ADV E1 SM-Bank 9

Solve $\log_2(6-x) - \log_2(4-x) = 2$ for x, where $x < 4$. *(2 marks)*

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25. L&E, 2ADV E1 SM-Bank 10

Solve the equation $2^{3x-3} = 8^{2-x}$ for x. *(2 marks)*

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26. Calculus, 2ADV C4 EQ-Bank 2

The population, D, of Tasmanian Devils in a sanctuary is given by $D(t)$, where t is the time in years after the sanctuary was established.

The devil population changes at a rate modelled by the function $\dfrac{dD}{dt} = 28e^{0.35t}$.

Calculate the increase in the number of Tasmanian Devils at the end of the first 8 years. Give your answer correct to two significant figures. *(3 marks)*

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27. Functions, 2ADV F2 SM-Bank 13

i. Show that the function $y = \dfrac{1-e^x}{1+e^x}$ is an odd function? *(1 mark)*

ii. Sketch $y = \dfrac{1-e^x}{1+e^x}$, labelling all intercepts and asymptotes. *(2 marks)*

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28. L&E, 2ADV E1 2009 HSC 1f

Solve the equation $\ln x = 2$. Give you answer correct to four decimal places. *(2 marks)*

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29. L&E, 2ADV E1 2010 HSC 4d

Let $f(x) = 1 + e^x$.

Show that $f(x) \times f(-x) = f(x) + f(-x)$. *(2 marks)*

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30. L&E, 2ADV E1 2004 HSC 6a

Solve the following equation for x:

$$e^{2x} + 3e^x - 10 = 0. \quad \textit{(2 marks)}$$

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31. L&E, 2ADV E1 2007 HSC 6a

Solve the following equation for x:

$$2e^{2x} - e^x = 0. \quad \textit{(2 marks)}$$

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32. Algebra, STD2 A4 2004 HSC 26a

i. The number of bacteria in a culture grows from 100 to 114 in one hour.
What is the percentage increase in the number of bacteria? ***(1 mark)***

ii. The bacteria continue to grow according to the formula $n = 100(1.14)^t$, where n is the number of bacteria after t hours.
What is the number of bacteria after 15 hours? ***(1 mark)***

Time in hours (t)	0	5	10	15
Number of bacteria (n)	100	193	371	?

iii. Use the values of n from $t = 0$ to $t = 15$ to draw a graph of $n = 100(1.14)^t$.
Use about half a page for your graph and mark a scale on each axis. ***(4 marks)***

iv. Using your graph or otherwise, estimate the time in hours for the number of bacteria to reach 300. ***(1 mark)***

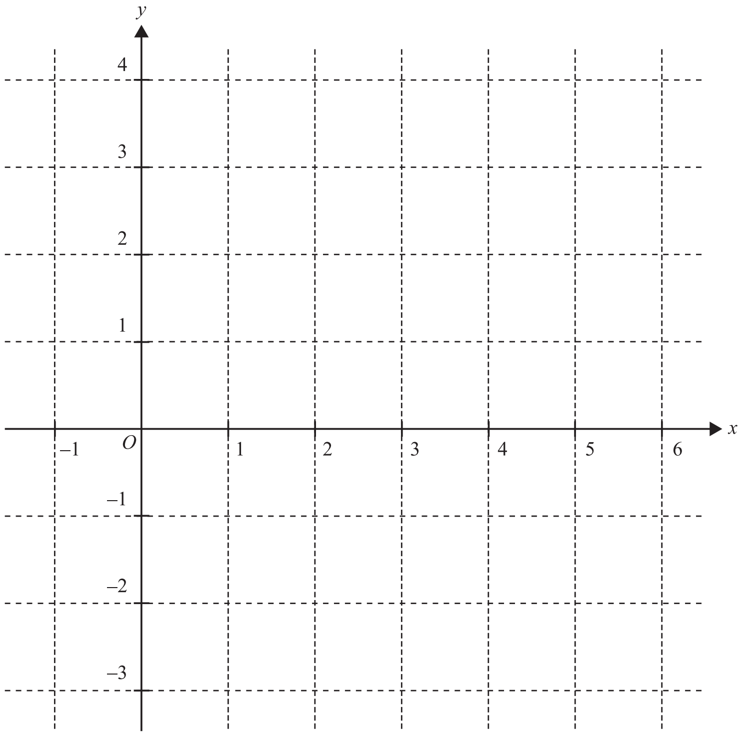
33. L&E, 2ADV E1 2019 MET1 4

Given the function $f(x) = \log_e(x-3)+2$,

a. State the domain and range of $f(x)$. ***(1 mark)***

b. **i.** Find the equation of the tangent to the graph of $f(x)$ at $(4, 2)$. ***(2 marks)***

ii. On the axes below, sketch the graph of the function $f(x)$, labelling any asymptote with its equation.
Also draw the tangent to the graph of $f(x)$ at $(4, 2)$. ***(4 marks)***



34. L&E, 2ADV E1 2008 HSC 7a

Solve $\log_e x - \dfrac{3}{\log_e x} = 2$ ***(3 marks)***

35. L&E, 2ADV E1 2019 HSC 15a

Solve $e^{2\ln x} = x + 6$ ***(2 marks)***

36. L&E, 2ADV E1 2016 HSC 14e

Write $\log 2+\log 4+\log 8+\ldots+\log 512$ in the form $a \log b$ where a and b are integers greater than 1. *(2 marks)*

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37. Algebra, STD2 A4 2012 HSC 30c

In 2010, the city of Thagoras modelled the predicted population of the city using the equation

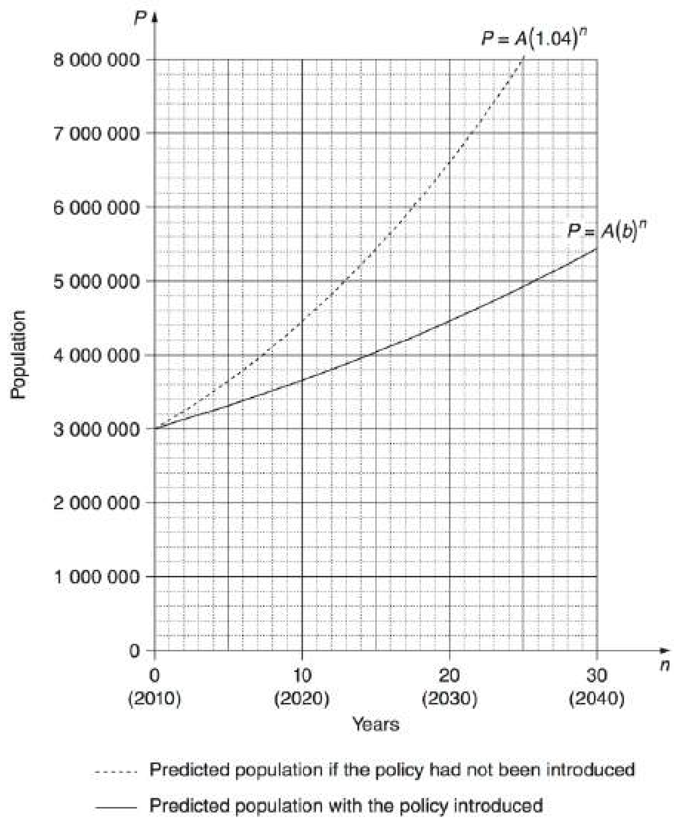
$$P=A(1.04)^{n}.$$

That year, the city introduced a policy to slow its population growth. The new predicted population was modelled using the equation

$$P=A(b)^{n}.$$

In both equations, P is the predicted population and n is the number of years after 2010.

The graph shows the two predicted populations.



(i) Use the graph to find the predicted population of Thagoras in 2030 if the population policy had NOT been introduced. *(1 mark)*

(ii) In each of the two equations given, the value of A is 3 000 000.
What does A represent? *(1 mark)*

(iii) The guess-and-check method is to be used to find the value of b, in $P=A(b)^{n}$.

(1) Explain, with or without calculations, why 1.05 is not a suitable first estimate for b. *(1 mark)*

(2) With $n = 20$ and $P = 4\,460\,000$, use the guess-and-check method and the equation $P = A(b)^n$ to estimate the value of b to two decimal places. Show at least TWO estimate values for b, including calculations and conclusions. *(2 marks)*

(iv) The city of Thagoras was aiming to have a population under 7 000 000 in 2050. Does the model indicate that the city will achieve this aim?
Justify your answer with suitable calculations. *(2 marks)*

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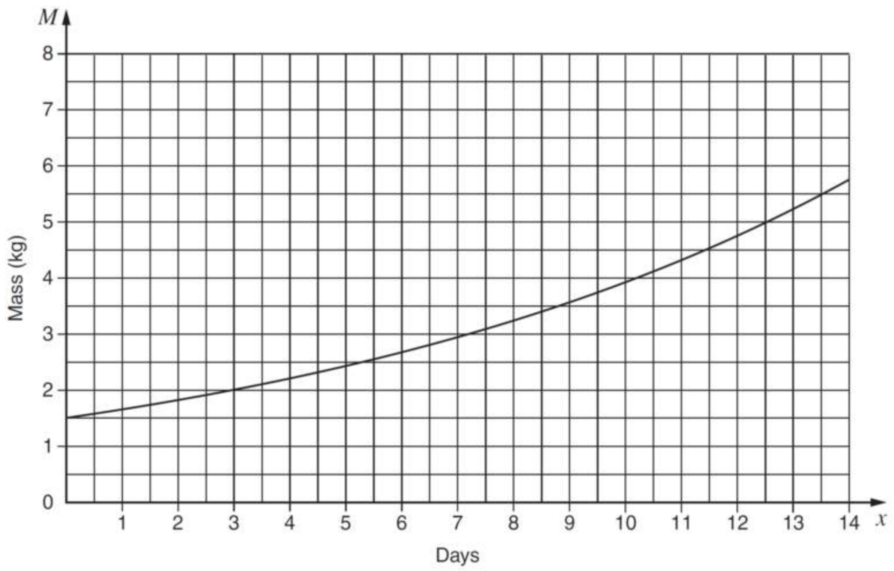
38. Algebra, STD2 A4 2016 HSC 29b

The mass M kg of a baby pig at age x days is given by $M = A(1.1)^x$ where A is a constant. The graph of this equation is shown.



(i) What is the value of A? *(1 mark)*

(ii) What is the daily growth rate of the pig's mass? Write your answer as a percentage. *(1 mark)*

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39. L&E, 2ADV E1 2019 MET1-N 4

Solve $\log_3(t) - \log_3(t^2 - 4) = -1$ for t. *(3 marks)*

Worked Solutions

1. Functions, 2ADV' F1 2019 HSC 1 MC

$$\begin{aligned} 4 - x &> 0 \\ -x &> -4 \\ x &< 4 \end{aligned}$$

$\Rightarrow A$

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2. L&E, 2ADV E1 SM-Bank 2 MC

$$\begin{aligned} y - 3 &= \log_a(7x - b) \\ a^{y-3} &= 7x - b \\ a^{y-3} + b &= 7x \\ \therefore x &= \frac{1}{7}(a^{y-3} + b) \end{aligned}$$

$\Rightarrow C$

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3. L&E, 2ADV E1 SM-Bank 4 MC

$$\begin{aligned} f(5x) &= 3\log_e(2(5x)) \\ \log_e(y) &= 3\log_e(10x) \\ &= \log_e(10x)^3 \\ y &= 1000x^3 \end{aligned}$$

$\Rightarrow D$

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4. Algebra, STD2 A4 SM-Bank 1 MC

From graph:

$$\begin{aligned} a^5 &= 4 \\ \therefore 2(a^5) &= 2 \times 4 \\ &= 8 \end{aligned}$$

$\Rightarrow$ B

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5. L&E, 2ADV E1 2012 HSC 7 MC

$$\begin{aligned}\log_e\left(a^2\right) &= \log_e\left(e^x\right)^2\\ &= \log_e\left(e^{2x}\right)\\ &= 2x\log_e e\\ &= 2x\end{aligned}$$

$\Rightarrow C$

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6. L&E, 2ADV E1 2013 HSC 9 MC

$$\begin{aligned}\text{Given } 5^x &= 4\\ \log_e 5^x &= \log_e 4\\ x\times\log_e 5 &= \log_e 4\\ \therefore x &= \frac{\log_e 4}{\log_e 5}\end{aligned}$$

$\Rightarrow C$

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7. L&E, 2ADV E1 2014 HSC 3 MC

$$\begin{aligned}\log_2(x-1) &= 8\\ x-1 &= 2^8\\ x &= 257\end{aligned}$$

$\Rightarrow D$

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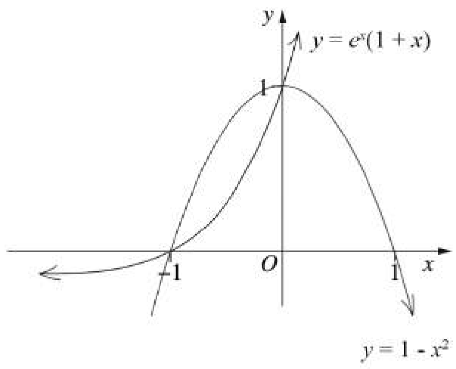
8. L&E, 2ADV E1 2015 HSC 8 MC



The graphs intersect at 2 points

$\therefore e^x(1+x) = 1-x^2$ has 2 solutions

$\Rightarrow C$

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9. L&E, 2ADV E1 2017 HSC 5 MC

$$\ln a = \ln b - \ln c$$

$$\ln a = \ln\left(\frac{b}{c}\right)$$

$$\therefore a = \frac{b}{c}$$

$\Rightarrow B$

Mean mark 51%.
COMMENT: Use of log laws here proved difficult for many students.

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10. L&E, 2ADV E1 2019 HSC 3 MC

$$\begin{aligned}\frac{a^2a^{-3}}{a^{\frac{1}{2}}} &= a^{-1}\times a^{-\frac{1}{2}}\\ &= a^{-\frac{3}{2}}\end{aligned}$$

$\Rightarrow B$

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11. L&E, 2ADV E1 2019 HSC 5 MC

$$\begin{aligned}\frac{\log_2 9}{\log_2 3} &= \frac{\log_2 3^2}{\log_2 3}\\ &= \frac{2\log_2 3}{\log_2 3}\\ &= 2\end{aligned}$$

$\Rightarrow\ A$

12. L&E, 2ADV E1 SM-Bank 11 MC

$e^{4x} - 5e^{2x} + 4 = 0$

Let $X = e^{2x}$

$$\begin{aligned}X^2 - 5X + 4 &= 0\\ (X-4)(X-1) &= 0\\ X &= 4 \text{ or } 1\end{aligned}$$

$$\begin{aligned}\therefore e^{2x} &= 4 & e^{2x} &= 1\\ 2x &= \log_e 4 & x &= 0\\ x &= \frac{2\log_e 2}{2}\\ &= \log_e 2\end{aligned}$$

$\Rightarrow\ C$

13. L&E, 2ADV E1 SM-Bank 6 MC

Solution 1

Using Change of Base:

$$\begin{aligned}&\log_c(a) + \log_a(b) + \log_b(c)\\ &\quad= \frac{\log_a(a)}{\log_a(c)} + \frac{\log_b(b)}{\log_b(a)} + \frac{\log_c(c)}{\log_c(b)}\\ &\quad= \frac{1}{\log_a(c)} + \frac{1}{\log_b(a)} + \frac{1}{\log_c(b)}\end{aligned}$$

$\Rightarrow\ B$

Solution 2

Let $x = \log_c(a)$

$$\begin{aligned}c^x &= a\\ x\log_a c &= \log_a a\\ x &= \frac{1}{\log_a c}\end{aligned}$$

Apply similarly for the other terms.

$\Rightarrow\ B$

14. L&E, 2ADV E1 2006 HSC 1a

$$\begin{aligned}e^{-0.5} &= 0.6065\ldots\\ &= 0.607\ \ \text{(to 3 d.p.)}\end{aligned}$$

15. Functions, 2ADV F1 SM-Bank 53

i. $\dfrac{1}{\sqrt[3]{7+\pi}} = (7+\pi)^{-\frac{1}{3}}$

ii. $\dfrac{1}{\sqrt[3]{7+\pi}} = 0.4619...$

$\quad = 0.462\ \ \text{(to 3 sig. fig.)}$

16. L&E, 2ADV E1 SM-Bank 8

$$3^{-4x} = \left(3^2\right)^{6-x}$$
$$3^{-4x} = 3^{12-2x}$$
$$-4x = 12 - 2x$$
$$2x = -12$$
$$\therefore x = -6$$

17. L&E, 2ADV E1 SM-Bank 12

Simplify using log laws:

$$\log_e(6x+10) = 2$$
$$6x + 10 = e^2$$
$$\therefore x = \frac{e^2 - 10}{6}$$

18. L&E, 2ADV E1 2011 HSC 1c

$$2^{2x+1} = 32$$
$$2^{2x+1} = 2^5$$
$$2x + 1 = 5$$
$$\therefore x = 2$$

MARKER'S COMMENT: Many students also correctly solved this by taking the logarithms of both sides.

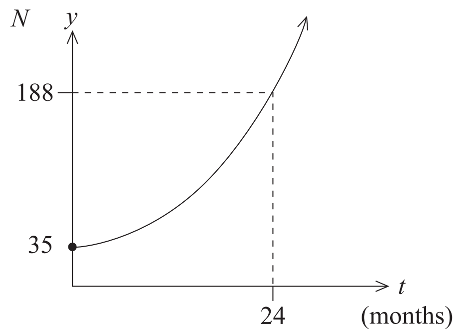
19. L&E, 2ADV E1 SM-Bank 2

i. $N = 35e^{0.07t}$

Find N when $t = 24$:

$$N = 35e^{0.07\times 24}$$
$$= 35e^{1.68}$$
$$= 187.79\ldots$$
$$= 188 \text{ birds}$$

ii.

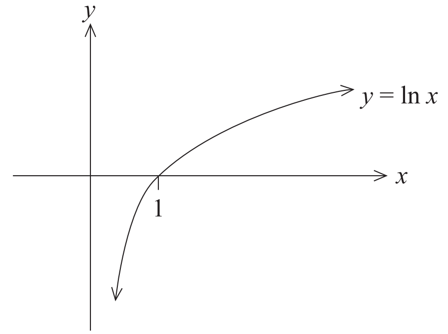


20. L&E, 2ADV E1 2005 HSC 5a

$$\log_3 7 = \frac{\log_e 7}{\log_e 3}$$
$$= 1.771\ldots$$
$$= 1.77 \text{ (to 2 d.p.)}$$

21. Functions, 2ADV F2 SM-Bank 1

i.

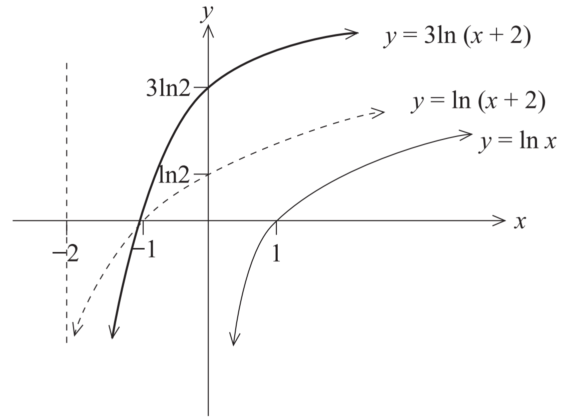


ii. Transforming $y = \ln x \Rightarrow y = \ln(x+2)$

$y = \ln x \quad \Rightarrow$ shift 2 units to left.

Transforming $y = \ln(x+2)$ to $y = 3\ln(x+2)$

$\Rightarrow$ increase each y value by a factor of 3



22. L&E, 2ADV E1 SM-Bank 14

i. $$\begin{aligned} f(0) &= \frac{8000}{1+1000e^0} \\ &= \frac{8000}{1001} \\ &= 7.99\ldots \end{aligned}$$

ii. $$\begin{aligned} f(365) &= \frac{8000}{1+1000e^{-0.12\times 365}} \\ &= \frac{8000}{1+1000e^{-43.8}} \\ &\approx 8000 \end{aligned}$$

After 1 year, the model predicts the total number of people infected by the virus is 8000.

23. L&E, 2ADV E1 SM-Bank 7

$$\begin{aligned} \log_3(5)^2 - \log_3(2) + \log_3(x) &= 2 \\ \log_3(25x) - \log_3(2) &= 2 \\ \log_3\left(\frac{25x}{2}\right) &= 2 \\ \frac{25x}{2} &= 3^2 \\ \therefore x &= \frac{18}{25} \end{aligned}$$

24. L&E, 2ADV E1 SM-Bank 9

Simplify using log laws:

$$\log_2\left(\frac{6-x}{4-x}\right)=2$$

$$\begin{aligned}2^2&=\frac{6-x}{4-x}\\16-4x&=6-x\\3x&=10\\\therefore x&=\frac{10}{3}\end{aligned}$$

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25. L&E, 2ADV E1 SM-Bank 10

$$\begin{aligned}2^{3x-3}&=2^{3(2-x)}\\3x-3&=6-3x\\6x&=9\\\therefore x&=\frac{3}{2}\end{aligned}$$

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26. Calculus, 2ADV C4 EQ-Bank 2

$$\begin{aligned}\int_0^8 28e^{0.35t}&=\left[28\times\frac{1}{0.35}e^{0.35t}\right]_0^8\\&=80\left(e^{0.35\times8}-e^{\circ}\right)\\&=80(16.44...-1)\\&=1235.57...\\&=1240\text{ (to 3 sig. fig.)}\end{aligned}$$

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27. Functions, 2ADV F2 SM-Bank 13

i. $f(x)=\dfrac{1-e^x}{1+e^x}$

$$\begin{aligned}f(-x)&=\frac{1-e^{-x}}{1+e^{-x}}\times\frac{e^x}{e^x}\\&=\frac{e^x-1}{e^x+1}\\&=-\frac{1-e^x}{1+e^x}\\&=-f(x)\end{aligned}$$

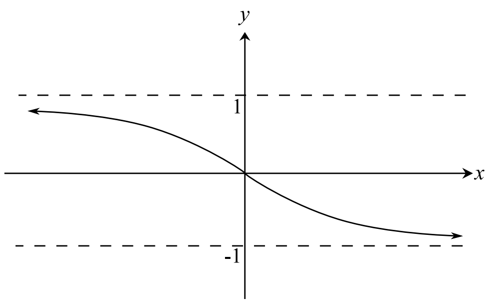
$\therefore f(x)$ is ODD.

ii. $y=\dfrac{1-e^x}{1+e^x}\times\dfrac{e^{-x}}{e^{-x}}=\dfrac{e^{-x}-1}{e^{-x}+1}=1-\dfrac{2}{e^{-x}+1}$

As $x\to\infty,\ \dfrac{2}{e^{-x}+1}\to2,\ y\to-1$

As $x\to-\infty,\ \dfrac{2}{e^{-x}+1}\to0,\ y\to1$

When $x=0,\ y=0$



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28. L&E, 2ADV E1 2009 HSC 1f

$$\begin{aligned} \ln x &= 2 \\ \log_e x &= 2 \\ x &= e^2 \\ &= 7.38905... \\ &= 7.3891 \ \ \text{(to 4 d.p.)} \end{aligned}$$

MARKER'S COMMENT: Students are reminded to write answers to more decimal places than required before rounding up.

29. L&E, 2ADV E1 2010 HSC 4d

$$\begin{aligned} &f(x) \times f(-x) \\ &\quad = (1+e^x)(1+e^{-x}) \\ &\quad = 1 + e^{-x} + e^x + e^x e^{-x} \\ &\quad = e^x + e^{-x} + 2 \end{aligned}$$

MARKER'S COMMENT: A common error in this question was not to realise that $e^x e^{-x} = e^0 = 1$.

$$\begin{aligned} &f(x) + f(-x) \\ &\quad = 1 + e^x + 1 + e^{-x} \\ &\quad = e^x + e^{-x} + 2 \\ &\quad = f(x) \times f(-x) \ \ \text{... as required} \end{aligned}$$

30. L&E, 2ADV E1 2004 HSC 6a

$$\begin{aligned} e^{2x} + 3e^x - 10 &= 0 \\ \therefore (e^x)^2 + 3e^x - 10 &= 0 \end{aligned}$$

$$\text{Let } \ X = e^x,$$

$$\begin{aligned} \therefore X^2 + 3X - 10 &= 0 \\ (X+5)(X-2) &= 0 \\ \therefore X &= 2 \ \text{ or } \ -5 \end{aligned}$$

$$\begin{aligned} \text{If } \ X &= 2 \\ e^x &= 2 \\ \ln e^x &= \ln 2 \\ x &= \ln 2 \\ \text{If } \ X &= \ -5 \\ e^x &= \ -5 \ \ \text{(no solution)} \end{aligned}$$

$$\therefore x = \ln 2$$

31. L&E, 2ADV E1 2007 HSC 6a

Solution 1

$2e^{2x} - e^x = 0$

Let $X = e^x$

$$2X^2 - X = 0$$
$$X(2X - 1) = 0$$
$$X = 0 \text{ or } \frac{1}{2}$$

When $e^x = 0 \ \Rightarrow$ no solution

When $e^x = \dfrac{1}{2}$

$$\ln e^x = \ln \frac{1}{2}$$
$$\therefore x = \ln \frac{1}{2}$$

Solution 2

$$2e^{2x} - e^x = 0$$
$$2e^{2x} = e^x$$
$$\ln 2e^{2x} = \ln e^x$$
$$\ln 2 + \ln e^{2x} = x$$
$$\ln 2 + 2x = x$$
$$x = -\ln 2$$
$$= \ln \frac{1}{2}$$

32. Algebra, STD2 A4 2004 HSC 26a

i. Percentage increase

$$= \frac{114 - 100}{100} \times 100$$
$$= 14\%$$

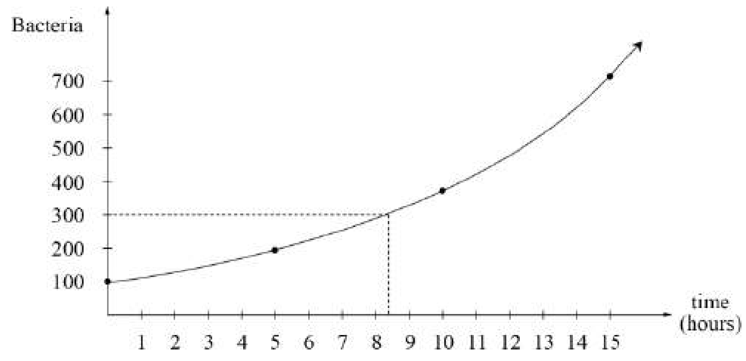
COMMENT: Common ADV/STD2 content in new syllabus.

ii. $n = 100(1.14)^t$

When $t = 15$,

$$n = 100(1.14)^{15}$$
$$= 713.793\ldots$$
$$= 714 \quad \text{(nearest whole)}$$

iii.



iv. Using the graph

The number of bacteria reaches 300 after approximately 8.4 hours.

33. L&E, 2ADV E1 2019 MET1 4

a. $\text{Domain}: x > 3$

$\text{Range}: y \in R$

b.i. $g(x) = \log_e(x-3)+2$

$$g'(x) = \frac{1}{x-3}$$

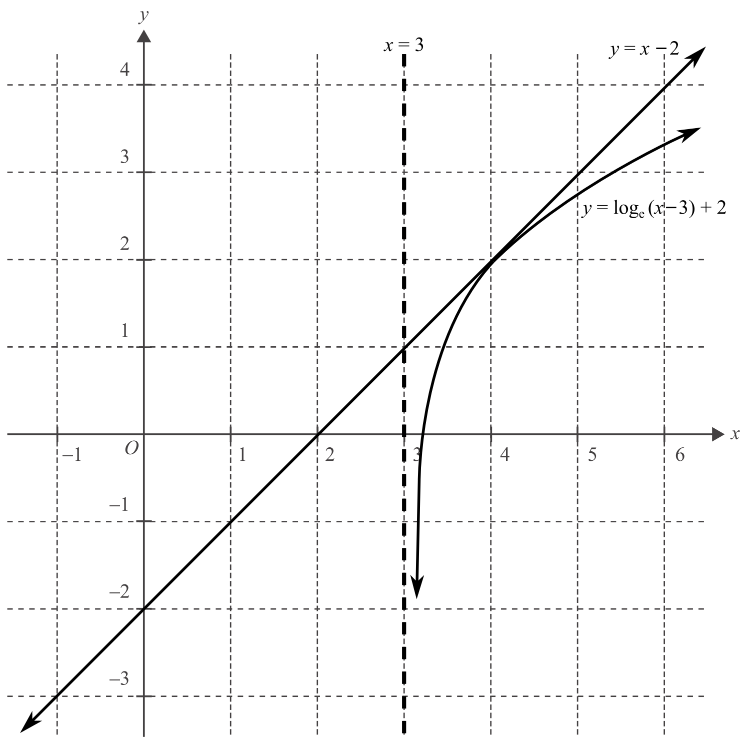
$$g'(4) = 1$$

$\text{Equation of tangent, } m = 1 \text{ through } (4,2):$

$$y - 2 = 1(x-4)$$

$$y = x - 2$$

b.ii.



34. L&E, 2ADV E1 2008 HSC 7a

$$\log_e x - \frac{3}{\log_e x} = 2$$

$$(\log_e x)^2 - 3 = 2\log_e x$$

$$(\log_e x)^2 - 2\log_e x - 3 = 0$$

$\text{Let } X = \log_e x$

$$\therefore\ X^2 - 2X - 3 = 0$$

$$(X-3)(X+1) = 0$$

$$X = 3 \qquad X = -1$$

$$\log_e x = 3 \qquad \log_e x = -1$$

$$x = e^3 \qquad x = e^{-1}$$

$$\therefore x = e^3 \text{ or } e^{-1}$$

IMPORTANT: Students should recognise this equation as a quadratic, and the best responses substituted $\log_e x$ with a variable such as X.

MARKER'S COMMENT: Many responses incorrectly stated that there is no solution to $\log_e x = -1$ or could not find x given $\log_e x = 3$.

35. L&E, 2ADV E1 2019 HSC 15a

$$e^{2\ln x} = x + 6$$

$$\ln e^{2\ln x} = \ln(x+6)$$

$$2\ln x = \ln(x+6)$$

$$\ln x^2 = \ln(x+6)$$

$$x^2 = x + 6$$

$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$\therefore x = 3 \ \ (x > 0)$$

♦ *Mean mark 47%.*

36. L&E, 2ADV E1 2016 HSC 14e

$$\log 2 + \log 4 + \log 8 + \ldots + \log 512$$

$$= \log 2^1 + \log 2^2 + \log 2^3 + \ldots + \log 2^9$$

$$= \log 2 + 2\log 2 + 3\log 2 + \ldots + 9\log 2$$

$$= 45\log 2$$

♦ *Mean mark 40%.*

37. Algebra, STD2 A4 2012 HSC 30c

(i) 2030 occurs at $n = 20$ on the x-axis.

Expected population (no policy) $= 6\,600\,000$

COMMENT: Common ADV/STD2 content in new syllabus.

(ii) A represents the population when $n = 0$ which is the population in 2010.

♦ *Mean mark part (ii) 48%*

(iii)(1) $P = A(1.05)^n$ would be steeper and lie above $P = A(1.04)^n$ since $1.05 > 1.04$

(iii)(2) Let $b = 1.03$

$$\begin{aligned} P &= 3\,000\,000 \times 1.03^{20} \\ &= 5\,418\,000 \end{aligned}$$

Let $b = 1.02$

$$\begin{aligned} P &= 3\,000\,000 \times 1.02^{20} \\ &= 4\,457\,800 \end{aligned}$$

$\therefore b = 1.02$

(iv) In 2050, $n = 40$

$$\begin{aligned} P &= 3\,000\,000 \times 1.02^{40} \\ &= 6\,624\,119 \ \text{(nearest whole)} \end{aligned}$$

Since the population is below 7 million, the model will achieve the aim.

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38. Algebra, STD2 A4 2016 HSC 29b

(i) When $x = 0$,

$$1.5 = A(1.1)^0$$

$\therefore A = 1.5$ kg

♦ *Mean mark 48%.*
COMMENT: Common ADV/STD2 content in new syllabus.

(ii) Daily growth rate

$$\begin{aligned} &= 0.1 \\ &= 10\% \end{aligned}$$

♦♦♦ *Mean mark part (ii) 6%.*
MARKER'S COMMENT: Interpretation of the exponential was very poorly understood.

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39. L&E, 2ADV E1 2019 MET1-N 4

$$\begin{aligned} \log_3(t) - \log_3(t^2 - 4) &= -1 \\ \log_3\left(\frac{t}{t^2 - 4}\right) &= -1 \\ \frac{t}{t^2 - 4} &= \frac{1}{3} \\ t^2 - 4 &= 3t \\ t^2 - 3t - 4 &= 0 \\ (t - 4)(t + 1) &= 0 \end{aligned}$$

$\therefore t = 4 \quad (t > 0,\ t \neq -1)$